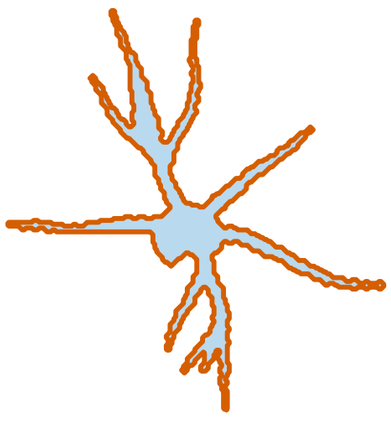
MicrogliaMorphology measurements, illustrated

27 features per cell from a synthetic ramified microglia. Colours link each name to its mark on the cell.

Cell shape (9)

ImageJ Measure on the thresholded single-cell mask

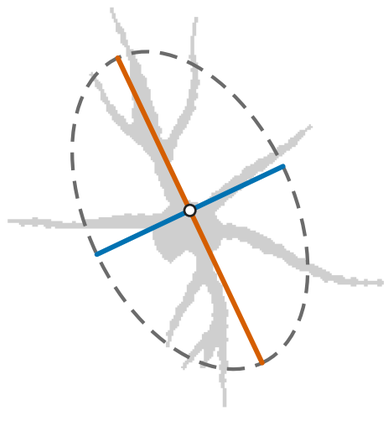
Area and perimeter



Area
Pixels inside the cell outline, in calibrated units (µm²). Shaded blue.

Perim.
Length of the traced outline around the cell, including every branch. Red line.

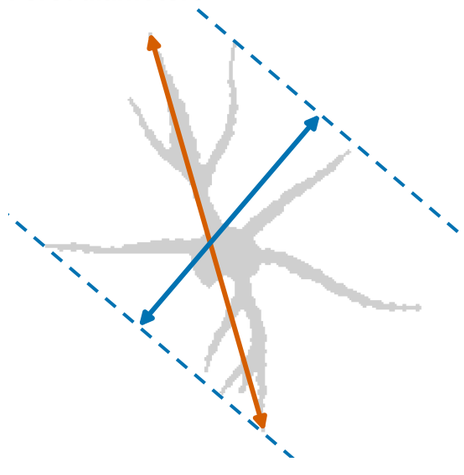
Major and minor axes



Major
Long axis of the best-fit ellipse, which has the same area, centre and orientation as the cell.

Minor
Short axis of that ellipse. It describes overall spread, not individual branches.

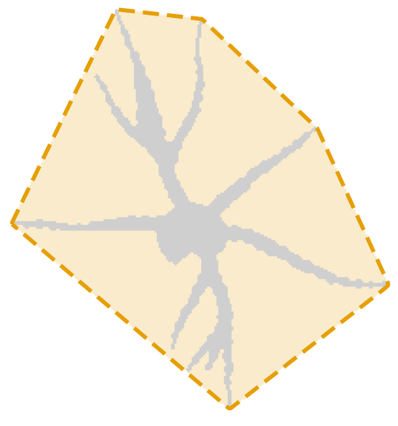
Feret diameters



Feret
Longest distance between any two points on the outline (maximum caliper diameter).

MinFeret
Smallest caliper width: the shortest gap between two parallel lines that just enclose the cell.

Shape ratios



Circ.
4π × Area / Perim². 1 is a perfect circle; long, branched outlines score near 0.

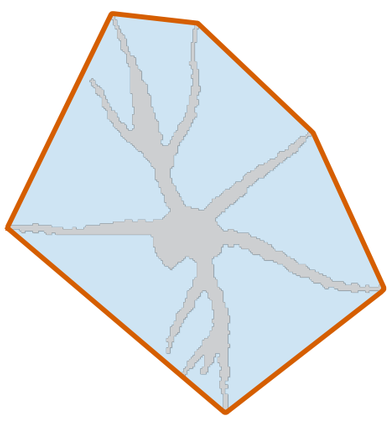
Round
4 × Area / (π × Major²). 1 is a circle; falls as the fitted ellipse elongates.

Solidity
Area / convex hull area. Orange gaps are the hull space the cell does not fill.

Convex hull (6)

The same measures applied to the convex hull, the smallest convex polygon around the cell

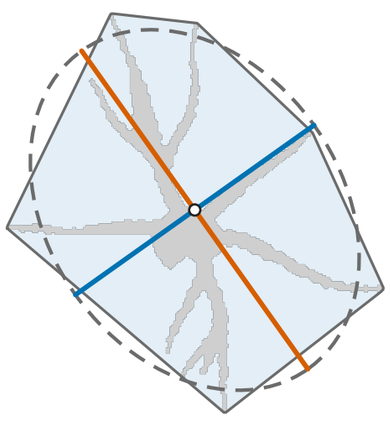
Hull area and perimeter



hull_Area
Area enclosed by the hull (blue). Always at least as large as Area.

hull_Perim.
Length of the hull outline (red). It ignores gaps between branches.

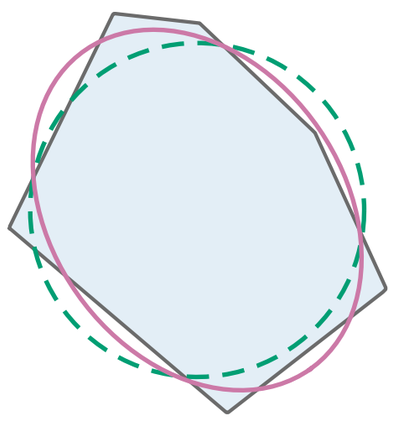
Hull major and minor axes



hull_Major
Long axis of the best-fit ellipse for the hull polygon.

hull_Minor
Short axis of that ellipse.

Hull circularity and roundness



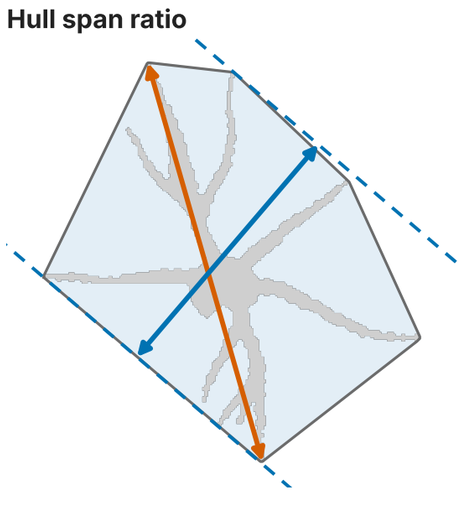
hull_Circ.
4π × hull_Area / hull_Perim². Compares the hull with a circle of equal area (green dashes).

hull_Round
4 × hull_Area / (π × hull_Major²). Compares the hull with its fitted ellipse (pink).

Hull-derived (3)

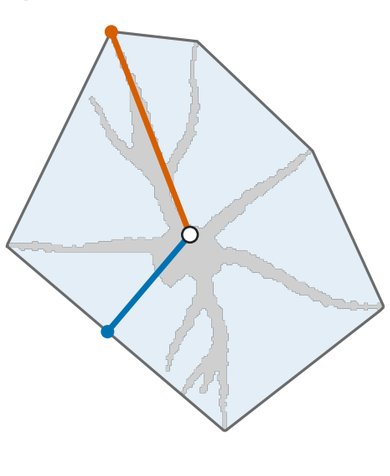
Calculated in the macro from the hull and the cell's centre of mass

Hull span ratio



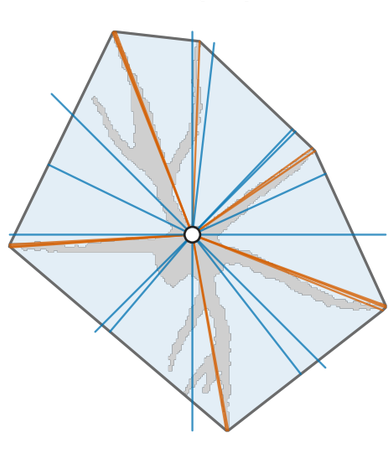
Hull span ratio
Major Feret / minor Feret of the hull (red / blue), the same calipers as Feret and MinFeret. 1 means the hull is equally wide in every direction; elongated or rod-like cells score higher.

Max/min radii from hull



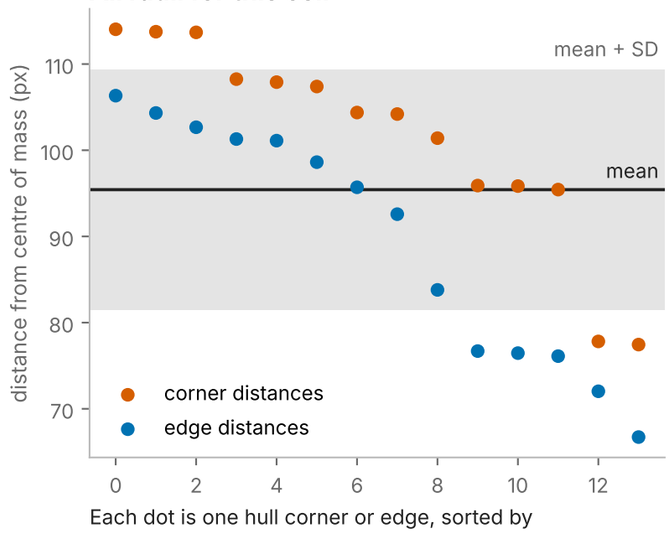
Max/min radii from hull
Longest distance from the centre of mass (white dot) to a hull corner (red), divided by the shortest distance to a hull edge (blue). High values flag an off-centre or lopsided hull.

Relative variation (CV)



Relative variation (CV) hull
Standard deviation / mean of all corner distances (red) and edge distances (blue) from the centre of mass. Higher values mean a more uneven hull.

All radii for this cell

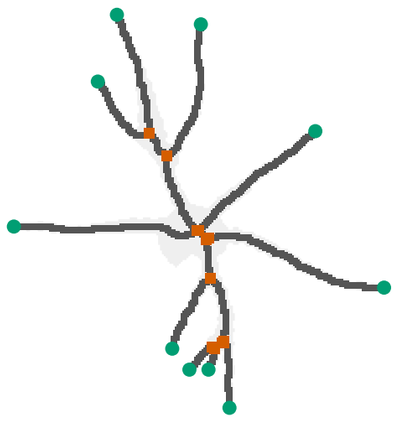


Each dot is one hull corner or edge, sorted by distance. The spread of the dots about the mean sets the CV; the top red dot over the lowest blue dot gives the max/min ratio.

Skeleton (9)

Skeletonize (2D/3D) followed by Analyze Skeleton (2D/3D); voxels are pixels in 2D

Voxel types

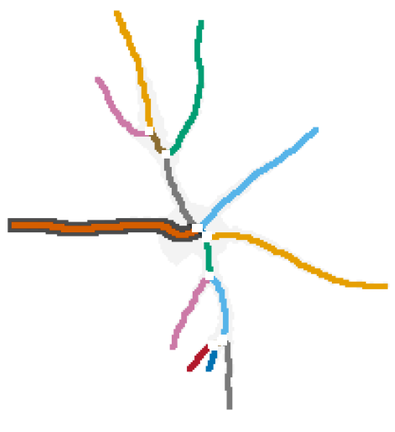


End-point voxels
Skeleton voxels with one neighbour: the tips of processes (green).

Slab voxels
Voxels with two neighbours: the body of a branch (grey).

Junction voxels
Voxels with three or more neighbours (red).

Branches

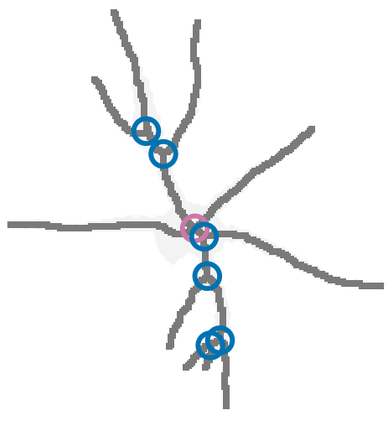


Branches
Skeleton segments between end points and junctions, one colour each (16 here).

Average Branch Length
Mean length of those segments, in calibrated units.

Maximum Branch Length
Length of the longest segment (outlined in red).

Junctions

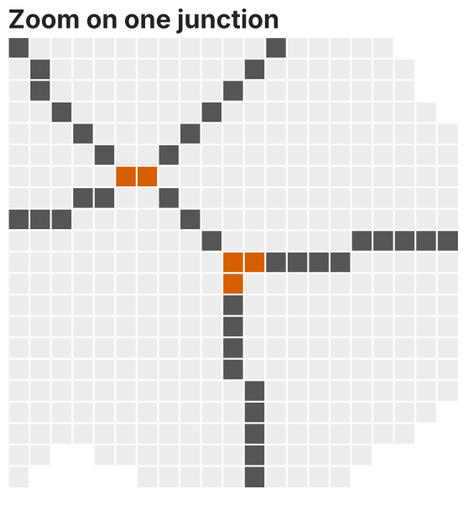


Junctions
Clusters of adjacent junction voxels, each counted once (7 circles here).

Triple points
Junctions where three branches meet (blue rings, 6).

Quadruple points
Junctions where four branches meet (pink rings, 1).

Zoom on one junction



How the voxels group
Each square is one skeleton voxel. The red voxels touching each other form one junction; the grey slab voxels leaving them are branches.